

A Four-Point-Line Construction for OEIS A172992

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Abstract

We verify a construction that places n rational points on four supporting lines. The correct parameter is $v = \lceil (s - d - 1)/2 \rceil$ (with d , not c). With this correction, for every $n \geq 28$ the construction guarantees $\lfloor (n^2 + 12)/28 \rfloor$ lines containing exactly four selected points, and a uniform dilation converts all coordinates to integers. In particular, it improves the known lower bounds for OEIS A172992 at $n = 59, 60$ to 124, 129. These are lower bounds, not proofs of optimality.

Construction and collinearity

Set

$$s = \left\lfloor \frac{3(n+1)}{7} \right\rfloor, \quad a = \left\lfloor \frac{s}{2} \right\rfloor, \quad b = \left\lceil \frac{s}{2} \right\rceil, \quad c = \left\lfloor \frac{n-s}{2} \right\rfloor, \quad d = \left\lceil \frac{n-s}{2} \right\rceil,$$

and

$$u = \left\lfloor \frac{s-c-1}{2} \right\rfloor, \quad v = \left\lceil \frac{s-d-1}{2} \right\rceil.$$

Define the integer intervals

$$\begin{aligned} A &= [0, a-1] \cap \mathbb{Z}, & B &= [0, b-1] \cap \mathbb{Z}, \\ C &= [u, u+c-1] \cap \mathbb{Z}, \\ D &= [-(b-1)+v, -(b-1)+v+d-1] \cap \mathbb{Z}. \end{aligned}$$

On $Y = 0, X = 0, Y = X, Y = -X$, respectively, place

$$\begin{aligned} P_x &= \left(\frac{1}{3n+x}, 0 \right), & Q_y &= \left(0, \frac{1}{n+y} \right), \\ R_z &= \left(\frac{1}{4n+z}, \frac{1}{4n+z} \right), & S_w &= \left(\frac{1}{2n+w}, -\frac{1}{2n+w} \right). \end{aligned}$$

There are $a + b + c + d = s + (n - s) = n$ points. The line through P_x, Q_y is

$$(3n+x)X + (n+y)Y = 1.$$

Substitution of R_z and S_w gives, respectively,

$$\frac{4n+x+y}{4n+z} = 1 \iff z = x+y, \quad \frac{2n+x-y}{2n+w} = 1 \iff w = x-y.$$

Thus, whenever $x \in A, y \in B, x+y \in C, x-y \in D$, the four points $P_x, Q_y, R_{x+y}, S_{x-y}$ are collinear. Such a line has a unique intersection with each supporting line, so it contains exactly these four selected points. Distinct pairs (x, y) give distinct lines because their two axis intercepts are different.

Counting

Put $L = -(b-1) + v$. For fixed $x \in A$, the admissible values of y form an integer interval. Hence the number of guaranteed four-point lines is

$$N(n) = \sum_{x=0}^{a-1} \left[\min\{b-1, u+c-1-x, x-L\} - \max\{0, u-x, x-L-d+1\} + 1 \right]_+, \quad (1)$$

where $[t]_+ = \max(t, 0)$. Formula (1) is also a direct, geometry-free way to check the count.

Write $n = 28q + r$, $0 \leq r < 28$. Substituting the endpoints in (1) and simplifying separately in the 28 residue classes gives

$$N(n) = 28q^2 + 2qr + h_r, \quad h_r = \left\lfloor \frac{r^2 + 12}{28} \right\rfloor.$$

For auditability, all values of h_r are displayed below; the calculation only compares the finitely many integer-interval endpoints in (1).

r	0	1	2	3	4	5	6	7	8	9	10	11	12	13
h_r	0	0	0	0	1	1	1	2	2	3	4	4	5	6
r	14	15	16	17	18	19	20	21	22	23	24	25	26	27
h_r	7	8	9	10	12	13	14	16	17	19	21	22	24	26

Consequently,

$$N(n) = \left\lfloor \frac{(28q+r)^2 + 12}{28} \right\rfloor = \left\lfloor \frac{n^2 + 12}{28} \right\rfloor, \quad n \geq 28.$$

Integer coordinates and the cases 59 and 60

All coordinates are rational. Let M be the least common multiple of all positive denominators in

$$\{3n + x : x \in A\} \cup \{n + y : y \in B\} \cup \{4n + z : z \in C\} \cup \{2n + w : w \in D\}.$$

The dilation $(X, Y) \mapsto (MX, MY)$ produces integer coordinates. Being invertible and linear, it preserves collinearity and the number of selected points on each line.

For $n = 59$ the parameters are $(s, a, b, c, d, u, v) = (25, 12, 13, 17, 17, 3, 4)$, and (1) gives 124. For $n = 60$ they are $(26, 13, 13, 17, 17, 4, 4)$, and (1) gives 129. As checked on September 6, 2026, the OEIS A172992 page still lists $a(59) = 123$, $a(60) = 126$ and notes that its terms are the best solutions known, not necessarily optimal ones.¹ The rigorous conclusion is therefore

$$A172992(59) \geq 124, \quad A172992(60) \geq 129.$$

If the sequence is updated according to its convention of recording the best known constructions, the candidate entries are 124 and 129; this construction does not rule out still larger values.

¹OEIS A172992, <https://oeis.org/A172992>, accessed September 6, 2026.